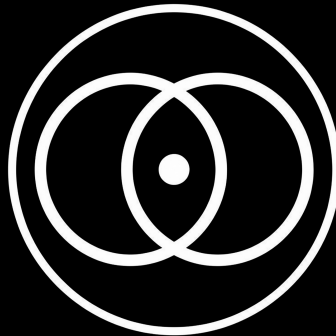




Metacognition: Know What You Know and What You Do Not — Engineering AI Self-Knowledge and Boundary Calibration

UDOS Reasoning Engine v7.5.0 · 2026-09-20

Evidence discipline: every UDOS number traces to the raw reports7/ JSON and is tagged verified / cpu-proto / cpu-proxy / unverified; single-seed runs are pilot point estimates, and ≥ 30 seeds are required before submission.

Working Title (EN): Engineering AI Self-Knowledge: Calibration, Abstention, and Identifiability

Prologue P0c of the flagship trilogy · together with P0 (Space Structure Is Itself Intelligence) and P0b (Structure + Scale Is Itself Intelligence) it forms the three axes “structure × scale × self-boundary” · UDOS reasoning engine v7.5.0 · 2026-09-20

The thesis in one sentence. Confucius said: “To know what you know and what you do not know, that is true knowledge.” Today’ s AI fails exactly here — its knowledge is nearly unbounded, yet it does not know where its own answers stop being reliable. This paper does not ask whether AI is conscious. It asks one engineering question: **how do we make a machine honestly say “I don’ t know” when it doesn’ t know?**

Evidence policy (stated once). Every UDOS number below comes from `reports7/*.json`, produced on an ordinary CPU with fixed random seeds and fully reproducible runs. Levels: `verified` (guarded by automated tests), `cpu-proto / cpu-proxy` (trends observed on a CPU prototype), `unverified` (external motivation only). Single-seed pilots are labeled as such; ≥ 30 seeds are required before submission. References were checked individually on 2026-09-20.

Abstract

The strongest language models today are, at bottom, supercharged probability matchers: given a question, the model finds the statistically most likely continuation in a vast parameter space and renders it in fluent prose. This mechanism has one fatal gap — **it has no concept of “I don’ t know.”** Confronted with a question outside its training data, it does not pause and acknowledge a boundary; it confidently produces a plausible-looking wrong answer. That is the root of hallucination: not a lack of knowledge, but a lack of metacognition — a sense of the boundaries of one’ s own knowledge.

This paper deliberately sets aside the metaphysical question of machine consciousness. Instead it reduces “self-knowledge” to three quantities that can be directly measured, acceptance-tested, and falsified:

1. **Calibration:** when the model says “I am 80% confident,” is it actually right 80% of the time?
2. **Selective abstention:** declining to answer on high-risk inputs, with the quality of refusal judged by a risk–coverage curve.
3. **Identifiability awareness:** knowing which latent variables cannot, in principle, be estimated under the current observation setup, and refusing to guess them.

We report three sets of direct evidence from a small world model of 967,796 parameters. First, split conformal prediction passes coverage at all three nominal levels, 80%/90%/95% (observed 0.9577 at the 95% level), including a “blind” path that cannot see the latent parameters. Second, estimating acceleration from a short 6-frame window yields an estimator whose error is 50.8% worse than simply predicting the mean (skill score -0.508) — the correct action is to abstain or

lengthen the observation window, not to force an estimate. Third, all 21 release gates pass; on contact tasks, direct control without a motion prior succeeds 0% of the time, rising to 100% once a gate routes those tasks to a hybrid policy.

These are single-seed preliminary results on a CPU prototype, not laws validated on real large models. The real contribution is a research program: replacing the unmeasurable word “wisdom” with three measurable engineering prerequisites, together with five experimentally falsifiable predictions. Until these three things are measurable, acceptance-tested, and falsifiable, any claim that a system “has wisdom” is not rigorous.

Keywords: metacognition; hallucination; calibration; selective prediction; conformal prediction; structural identifiability; world models; safety gating

1 A Car with a Huge Engine and No Brakes

1.1 Confucius as a diagnostic report

In the Analects (II.17), Confucius says: **“To know what you know and what you do not know, that is true knowledge.”** The sentence is not about the amount of knowledge; it is about awareness of one’s own epistemic state: knowing what you know is clarity; knowing what you do not know is wisdom.

Run today’s AI through that diagnostic and the result is striking. In sheer volume of knowledge, a large model has read more text than any human could encounter in a lifetime. In self-awareness it remains an infant: it does not know where its answers come from, how reliable they are, or where its knowledge ends.

One rule must be set first: **Confucius’ s sentence is a moral ideal (an ought), not scientific evidence (an is).** We cite it only to name the ancient intuition we are trying to engineer. Every scientific claim in this paper rests on the three measurable quantities of Section 2 and the experiments of Section 4. If our predictions are experimentally refuted, Confucius cannot save the paper.

1.2 A precise definition of hallucination

Hallucination is often defined loosely as “the model says something wrong.” That definition is too broad — every finite model errs. We adopt a sharper and more dangerous definition:

Hallucination is when a model outputs, for something it fundamentally does not know, an answer every bit as confident as its answer to something it does know.

The first (being wrong) is a competence deficit; the second (being confidently wrong) is a missing self-boundary. A self-aware system, when out of its depth, should emit a wide interval, abstain, or go retrieve information; a system without self-awareness still hands back a firm point estimate.

Why does a probability matcher lack a boundary by construction? Because its training objective — predict the next token — contains no dimension for “do I actually know this?” It is trained always to produce the most likely continuation, never to judge whether it should produce one at all.

1.3 What this paper does and does not do

This paper does **not** claim that AI has consciousness, a self, or subjective experience. Calibration, abstention, and identifiability awareness are all behavioral, engineering quantities; a machine can perform all of them with no inner experience whatsoever (Section 6 faces this objection head-on).

What the paper **does** is find three engineering prerequisites for “self-knowledge” and make them measurable, acceptance-tested, and falsifiable. We are not building the engine — ability is the subject of P0/P0b — we are installing the **brakes**. Brakes produce no power, but a brakeless car becomes more dangerous the more powerful its engine is.

2 Taking “Self-Knowledge” Apart into Three Measurable Things

“Knowing what you know” sounds mystical; taken apart, each piece has mature mathematical tooling.

2.1 Calibration: 80% confident should mean 80% correct

Calibration asks whether stated confidence matches actual correctness. A model that says “90% confident” on 100 questions and gets exactly 90 right is perfectly calibrated; if it gets 55 right, it is overconfident.

Standard tools are the Expected Calibration Error (ECE — bin samples by confidence and compare each bin’s mean confidence with its accuracy) and the Brier score (mean squared error between confidence and correctness). **The key point: calibration and accuracy are different things.** A 70%-accurate model can be well calibrated (it knows it is right only seven times in ten), while a 95%-accurate model can be badly calibrated (it always claims certainty). Guo et al. (2017) found that modern deep networks are systematically overconfident, and that a one-parameter post-processing step called temperature scaling substantially improves calibration with almost no change in accuracy — evidence that “knowing whether you are right” and “being right” are separable axes.

2.2 Selective abstention: permitting the words “I won’ t answer”

Selective classification adds an abstention switch: on inputs whose confidence falls below a threshold, the model does not answer and hands off to a human or another system.

Its central tool is the **risk–coverage curve**: coverage is the fraction of samples the model answers; risk is its error rate on the answered samples. Sweeping the confidence threshold from loose to tight lowers both coverage and risk, and the curve should sit as far toward the lower-left as

possible. A scalar summary is the area under this curve, AURC, where **smaller is better**. The framework goes back to Chow’ s (1957) optimum error–reject trade-off and was formalized by El-Yaniv & Wiener (2010) and Geifman & El-Yaniv (2017).

More abstention is not always better — in medicine or autonomous driving, “I don’ t know” can be more dangerous than a wrong answer when it hands control back to an already-overloaded human. A good system lets the operator make an explicit trade-off under a risk budget rather than simply refusing whenever uncertain.

2.3 Identifiability awareness: some things are unknowable under the current observations

The first two quantities ask “how sure am I of this answer?” The third is more fundamental: **can this quantity be estimated at all under the current observation structure?**

Systems identification (Bellman & Åström, 1970) distinguishes:

- **Observability**: can the internal state be recovered from an observation trajectory? A yes/no structural property.
- **Identifiability**: given the data, can the unknown parameters be estimated accurately enough? A question of “how well.”

A quantity can be highly observable (its shadow is present in the data) yet remain non-identifiable (the shadow is too weak or the sample too short, so the estimator is worse than not modeling at all). A **self-aware system must detect this and refuse to guess** — the piece this paper most wants to add to the classical calibration literature. We quantify it with a skill score:

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A positive score means the model beats the naïve mean; zero means it learned nothing; a **negative score means the model is worse than doing nothing — and the rational action is not to use it**.

2.4 The triad: a minimal engineering framework

Putting the three together, we describe an “engineering self-model” as a triad:

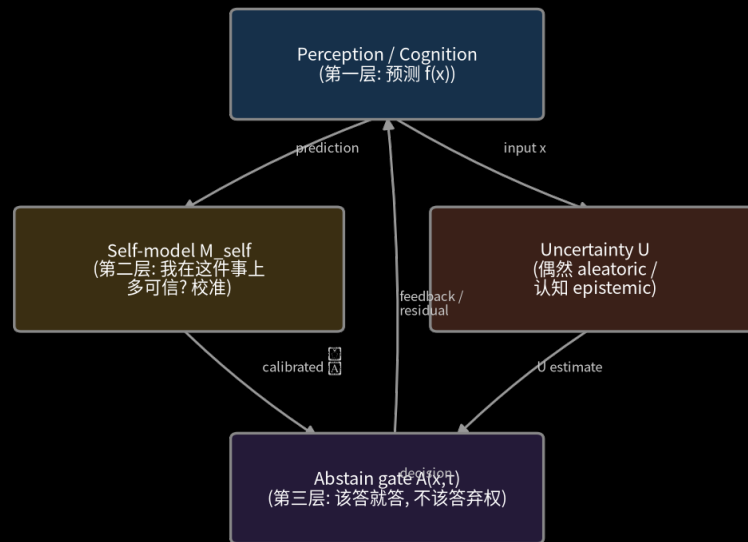
$(M_{\text{self}}, U, A(x, \tau))$

- M_{self} (self-model): the system’ s estimate of its own competence — how it has historically performed on this kind of task;
- U (uncertainty): the uncertainty attached to the current prediction, distinguishing **aleatoric uncertainty** (noise intrinsic to the data, irreducible with more data) from **epistemic uncertainty** (uncertainty from not having seen enough, removable by collecting more data);
- $A(x, \tau)$ (abstention policy): abstain, hand off, or call a tool when uncertainty exceeds threshold τ .

Uncertainty must also come with **coverage-guaranteed intervals**, not just points; conformal prediction provides them (Section 4.2).

A note on levels. Cognition can be coarsely layered: L1 perception (pattern recognition), L2 cognition (reasoning and planning), L3 metacognition (knowing what you do and do not know), L4 hyper-cognition (recalibrating your own calibration, even posing new questions). This paper does L3 only; L4 is open. The four-layer scheme is our organizing frame, not a consensus of the literature.

Fig.1 Metacognitive Triple: prediction — self-model/uncertainty — abstain gate (schematic)



The loop: $f(x)$ produces a prediction; M_self and U estimate how trustworthy it is; $A(x, t)$ decides to answer or abstain; residuals feed back to recalibrate M_self .

Figure 1: The metacognition loop. After acting, the system observes the outcome; the world model produces a prediction and an uncertainty; calibration/identifiability modules judge “do I know this?” ; the abstention policy decides whether to answer, refuse, or go external; the result is written back into the self-model.

3 The Road Already Traveled

This paper is not built in a vacuum. Four research lines have, over decades, shaped the three pieces.

The **concept of metacognition** comes from developmental psychology. Flavell (1979) coined the term and divided metacognition into metacognitive knowledge (understanding one’s own cognitive processes) and metacognitive monitoring (watching one’s current state of comprehension). We borrow the engineering counterpart of the latter: online monitoring of the reliability of one’s own predictions.

Calibration was rediscovered in the deep-learning era. Guo et al. (2017) showed modern networks are overconfident and that temperature scaling (dividing logits by a scalar T before softmax) improves calibration cheaply without changing predicted classes — direct support for prediction F1.

Abstention has a complete theoretical line: Chow (1957) introduced optimal recognition with a reject option; El-Yaniv & Wiener (2010) founded the risk-coverage framework for noise-free selective classification; Geifman & El-Yaniv (2017) built a selective classifier for any trained network; SelectiveNet (2019) baked the reject option into the network itself.

Distribution-free uncertainty quantification is supplied by conformal prediction. The Angelopoulos & Bates (2021) primer systematizes split conformal: compute a quantile on calibration data untouched by training, and prediction intervals inherit a coverage guarantee that assumes nothing about the distribution — only exchangeability between calibration and test samples.

Do large models know what they know? Kadavath et al. (2022) ran the key experiment: models' non-verbal ability to predict their own correctness in multiple-choice settings correlates with actual accuracy but is far from perfect, and RLHF fine-tuning harms calibration. Reflexion (Shinn et al., 2023) has agents write reflections into language memory. World Models (Ha & Schmidhuber, 2018) suggests another route: use prediction error itself as the signal for “I have not seen this.” Hendrycks & Gimpel (2017) found the maximum softmax value to be a cheap baseline for out-of-distribution detection.

Together these leave a gap: **the calibration literature rarely asks whether a quantity is structurally estimable; the identifiability literature rarely connects to LLM abstention engineering; conformal prediction is rarely placed inside a unified “metacognition” frame.** This paper joins the three pieces and runs them end-to-end on a controllable small world model.

4 What We Actually Measured

The platform is a small world model inside the UDOS engine: a 2-layer gated recurrent network, hidden size 256, 32-dimensional scene latent, 6-frame input windows, **967,796 parameters** in total. It learns the state-action-next-state transition in a synthetic 2-D dynamical world deliberately containing four regimes: uniform motion, acceleration, springs, and collisions.

4.1 The world model itself: first prove it can predict

From v7.0.1 to v7.0.3, overall prediction error (MSE) on the blind test set fell from 0.0459 to **0.01037**, roughly a $4.4\times$ improvement. By regime, collisions are easiest (0.00304), then uniform motion (0.00573), acceleration (0.01410), and springs hardest (0.01862). Median single-step prediction latency (batch size 1) is 2.36 ms; a 4-step rollout takes 4.83 ms — light enough for a real-time control loop. (Level verified.)

This step is the precondition for everything else: **a model that cannot predict cannot assess how good its predictions are.**

4.2 Conformal prediction: promise 95%, observe 95.8%

We wrap the world model's predictions in split-conformal intervals: 640 windows and 128 trajectories serve as a held-out calibration set; on fresh test trajectories we check how often the true value falls inside. Two paths are run: an oracle path allowed to see the true latent parameters, and a blind path that cannot (closer to real deployment).

Table 1 Conformal coverage: nominal vs observed (level verified, ± 0.05 tolerance)

Nominal	Oracle coverage	Oracle width	Blind coverage	Blind width
80%	0.7839	0.0430	0.7758	0.0504
90%	0.8984	0.0858	0.9001	0.1081
95%	0.9577	0.1612	0.9559	0.2052

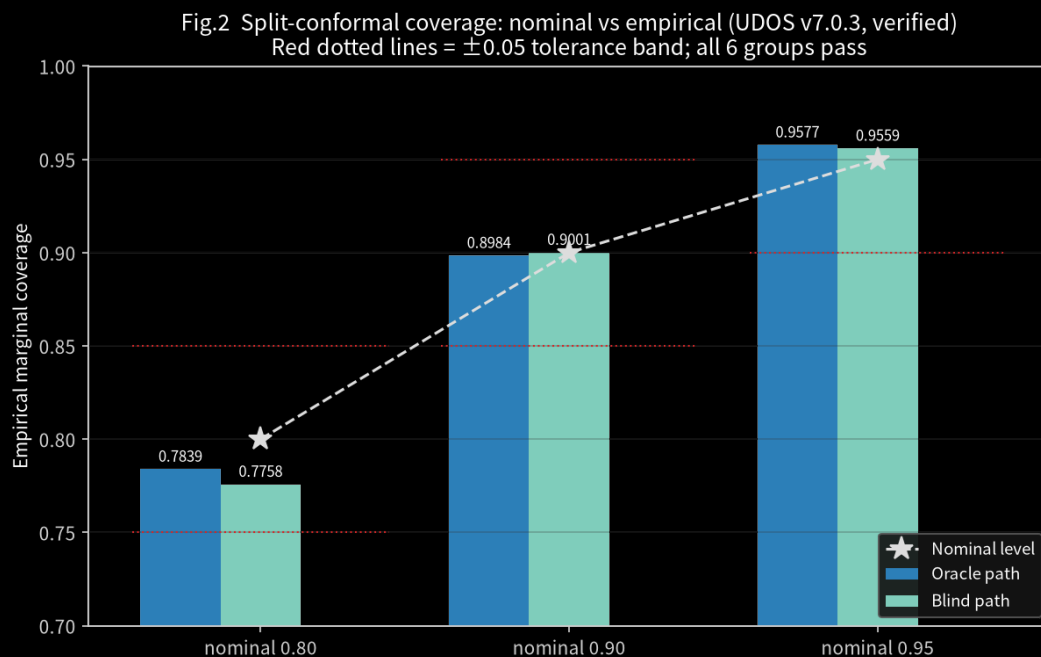


Figure 2: At all three nominal levels, observed coverage lies within ± 0.05 (all 21 release gates pass). The blind path, which cannot see latent parameters, still meets coverage, at the price of systematically wider intervals (0.2052 vs 0.1612 at 95%).

How to read this table. The model need not “feel” how certain it is; it simply measures a quantile once on calibration data and obtains intervals with a mathematical guarantee — “when I say 95%, roughly 95% of future true values fall in.” The wider blind intervals matter most: **with less information the system automatically becomes more humble — the germ of boundary awareness.**

The conformal guarantee has an explicit premise: calibration and test data must be exchangeable. Under distribution shift or out-of-distribution inputs the premise fails and coverage degrades — the subject of prediction F4, not a limitation we paper over.

4.3 The acceleration negative result: when guessing is worse than not guessing

This is the most counterintuitive and valuable result. We ask the estimator to recover four latent physical quantities from a 6-frame window, 160 samples each, and compute the skill score of Eq. (1):

Table 2 Identifiability of four latent quantities (n=480; main gate verified, breakdown cpu-proto)

Latent quantity	Estimator error	Mean-prediction error	Skill	Observability
Initial velocity v_0	0.6606	0.9683	+0.318	0.771
Angular frequency ω	0.1471	0.2563	+0.426	0.914
Collision restitution v_2	0.0432	0.2396	+0.820	0.908
Acceleration a (6-frame window)	0.9256	0.6138	-0.508	0.798

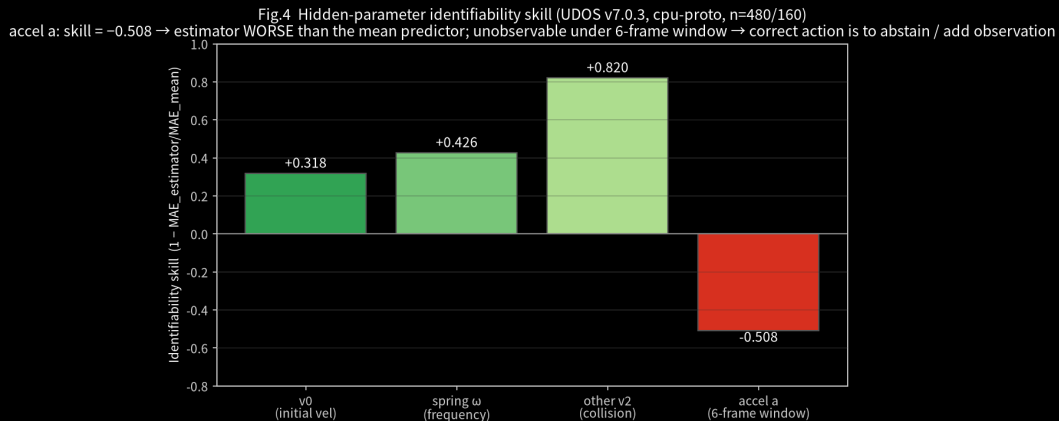


Figure 4: The first three quantities have positive skill — the model genuinely learns them. Acceleration under a 6-frame window scores -0.508: the estimator is 50.8% worse than predicting the mean. Note that its observability is not low (0.798) — acceleration's shadow is in the data, but six frames are too few to identify it.

The negative result says three things. First, high observability does not imply identifiability: the shadow of acceleration is present, but the window is too short for any model to estimate it. Second, a system without self-awareness would here confidently emit that terrible point estimate — a textbook hallucination; a self-aware system that sees negative skill should abstain or extend

the window from 6 to 12/24 frames. Third — and this is the core of prediction F2 — **this is a hard boundary of the observation structure, not something a bigger model routes around.**

4.4 Gating: wiring “I don’ t know” into real actions

Calibration and intervals produce no safety value if they stay in reports. We wire uncertainty into a set of pre-release gates, 21 checks in total, all passing (coverage tolerance ± 0.05 , parameter fan ± 0.10).

Two gates make the point most clearly:

- **Contact gate:** in embodied hybrid-control experiments (40 episodes per arm, level cpu-proxy), pure semantic control of the arm succeeds 75% with collision rate 0.25 and 1.0 takeover on average; motion-prior-only control succeeds 52.5%; on contact tasks, direct control without a motion prior succeeds 0% of the time. Once the gate routes contact tasks to the hybrid policy, it reaches **100% success**, zero collisions, and only 0.047 takeovers. **The gate does not make the model smarter; it stops it from forcing its way through tasks it is bad at.**
- **Kill switch:** when the retry budget is set to zero, the system honestly refuses the task, the governance check returns not-ok and it degrades — it does not fake completion to “finish the job.”

Gate trigger rates carry information too: uniform motion 63.4%, acceleration 66.2%, collisions 15.4%, springs 0%. The model calls for help more often on the dynamics it finds harder to identify, consistent with the ordering in Section 4.3.

4.5 The collective level: brakes for one agent and for a million

In multi-agent systems the self-boundary has a collective meaning: the whole swarm must also know whether it can still converge. Three results from companion papers:

- A Byzantine-fault-tolerant committee with $n=7$, $f=2$ decides all 24 work orders correctly (quorum $2f+1$);
- Under fault injection, a system with the Owner/Trace/Stop-Condition governance matrix converges 24/24 in both clean and stressed runs, and isolates 3 faulty agents through 5 reworks (**single-seed pilot; ≥ 30 seeds required**);
- The internal task market conserves budget, payment, and balance exactly ($240=240=240$); conservation is itself a global self-check that the system state remains consistent.

4.6 Evidence summary

Table 3 Mapping the three core evidence sets to the metacognition triad

Triad component	Experiment	Key number	Level
Uncertainty U (calibration)	Split conformal	0.9577 oracle / 0.9559 blind at 95%	verified

Triad component	Experiment	Key number	Level
Self-model M_{self} (identifiability awareness)	Four-parameter identification	Acceleration skill -0.508 $\rightarrow$ abstain	main gate verified
Abstention policy $A(x, \tau)$	21 gates	All pass; contact tasks $0\% \rightarrow 100\%$	verified/cpu-proxy

5 How This Paper Could Be Proved Wrong

We keep the three most decisive of five predictions, each with a minimal experiment and a decision rule.

F1 | Calibration and performance are separable axes. Minimal experiment: take one trained checkpoint (no retraining), freeze the weights, do a one-dimensional grid search over temperature T on a calibration set, and measure ECE and accuracy with 5-fold cross-validation. Decision rule: F1 holds if some T cuts ECE by $\geq 30\%$ relative while accuracy moves within noise ($< 0.1\%$); F1 dies if every measurable calibration gain comes with a proportional accuracy gain (calibration is a mere by-product of performance).

F2 | When the observation structure is insufficient, more parameters cannot rescue it. Minimal experiment: hold the 6-frame window fixed, only double and quadruple estimator capacity (hidden $256 \rightarrow 512 \rightarrow 1024$, layers $2 \rightarrow 4$), and measure acceleration skill; run a positive-control arm that extends the window to 12/24 frames. Decision rule: F1 holds if skill stays ≤ 0 as capacity grows but turns positive immediately with a longer window — cleanly separating observation structure from model capacity; F2 dies if a big model lifts skill stably above $+0.2$, meaning scale does break the structural boundary and the paper retreats to “a small-model constraint.” This is the boldest and most easily tested prediction.

F3 | Gating answers more under the same error budget. Minimal experiment: on one benchmark run two arms — bare model vs the same model plus conformal/confidence gating; fix a target error rate, sweep the threshold, compare coverage and AURC. Decision rule: F3 holds if the gated arm covers significantly more (significantly smaller AURC); F3 dies if the two risk-coverage curves coincide within statistical noise (abstention merely passes hard problems to someone else).

Two secondary predictions: **F4**, explicit uncertainty channels (conformal intervals, parameter fans) should degrade more slowly than verbalized confidence under out-of-distribution shift — Kadavath already found verbal confidence imperfect and harmed by RLHF; **F5**, governed multi-agent systems should converge more often under fault injection, but the current 24/24 is a single-seed point estimate and **must be rerun with ≥ 30 seeds with a 95% confidence interval**; if that interval contains zero, F5 dies.

6 Where We Might Be Wrong

First, calibration is not consciousness. Temperature scaling divides by a scalar; conformal prediction looks up a quantile; a gate compares two numbers — none of this requires the system to “reflect on its own reflection.” We fully accept the behaviorist objection and put it in the title: our “metacognition” is an engineering term denoting the three behavioral quantities, not Flavell’s metacognitive experience, let alone consciousness. A system can be perfectly calibrated with no subjective experience at all. Reducing self-knowledge from a question of inner experience to one of behavioral measurement is precisely what lets us build it.

Second, over-abstention is another kind of failure. A system that refuses at the slightest doubt is useless; in a doctor-scarce setting, handing the patient back, or handing control back to a distracted driver, can be worse than answering. Confidence can also be fooled by adversarial inputs, and conformal exchangeability fails out of distribution. Our AURC currently measures only “how much answered vs how much wrong,” not “how much refused and at what cost” — modeling the cost of refusal is explicit future work.

Third, introspection itself is unreliable. Asking a model to state its own confidence produces a self-report that itself needs calibrating; Reflexion-style verbal reflection can mis-attribute a failure and write the wrong cause into memory, contaminating the next trial. Hence our one-line stance: **do not trust what it says; trust what it measures.** Internal confidence signals may be candidates, but they must be calibrated against an independent calibration set.

Fourth, Confucius is not evidence, and a synthetic world is not the real world. The Analects only motivates; every UDOS number comes from a deterministic CPU prototype (point-mass dynamics, rule-based judges, a single seed) — not a real large model, not real hardware, with no 95% confidence intervals. The acceleration negative result was measured on an injected short-window defect; real-world “insufficient observation” is more complex. We therefore position this paper as **a research program plus preliminary silicon evidence**: it shows the mechanism’s shape works in a simplified world, not that its constants hold in real systems. Moving conformal prediction onto real LLM output spaces and rerunning every pilot with ≥ 30 seeds are the two things that must be done before submission.

7 Consciousness: Explicitly Out of Scope

This chapter marks the boundary. The paper nowhere **claims** consciousness, qualia, or a sense of self. Theories such as Attention Schema Theory (AST) or Global Workspace Theory (GWT) superficially resemble our self-model, but we **neither endorse, cite, nor test** them: they are hypotheses about human consciousness whose applicability to silicon is itself contested, while our M_{self} is merely a numeric estimate of the system’s own competence — not an “attention schema.”

A calibrated system need not be conscious; a conscious system need not be calibrated. The existence of this chapter is itself the discipline of “knowing what you do not know” : **this paper knows it does not know what consciousness is, and does not pretend to.**

8 Engineering Implications: Install the Brakes Before Flooring the Throttle

First, calibration should be a first-class acceptance criterion, not a post-launch patch. Temperature scaling is cheap, but only if you actually measure it. Writing ECE, coverage, and AURC into CI gates (as our 21 gates do) is far cheaper than firefighting afterward.

Second, hallucination is to be managed, not fantasized away. No finite system is right on every sample; what you can do is ensure that, when it does not know, it does one of three things: emit a wide interval (conformal), abstain (gating), or retrieve external knowledge (RAG/Toolformer). “Eliminate hallucination with a bigger model” works only in-distribution — the root of hallucination is a missing self-boundary, not insufficient scale, in direct correspondence with prediction F2 of P0b.

Third, identifiability must be considered at observation-design time. The acceleration result tells architects: if the window is too short or the sensors cannot see a key quantity, no larger model helps; the correct move is a longer window or better sensors, not more parameters.

Fourth, the trilogy is multiplicative. P0 says structure determines the shape of the competence function; P0b says scale determines where you stand on the curve; this paper says the self-boundary determines where you must stop. Structure and scale are the engine; the self-boundary is the brake: a brakeless car is more dangerous the more power it has, while a car with only brakes and no engine is just a roadblock. All three are required.

9 Conclusion

P0 says: space structure is itself intelligence. P0b says: structure plus scale is itself intelligence. This paper says: **knowing what one does not know is the third axis of intelligence.**

We have not proved that any system possesses wisdom. What we did is far more modest: we replaced the unmeasurable word “wisdom” with three measurable engineering prerequisites — calibration, abstention, identifiability awareness — ran them end-to-end on a 968k-parameter world model, and set up five targets that experiments can knock down.

AI’ s coming-of-age will not be the day it answers every question. It will be the day it first honestly says, “I don’ t know — let me look that up,” or “this lies beyond what my observations can determine.” The Analects held that ideal for 2,500 years; what this paper does is convert a moral precept into a measurable, acceptance-testable, falsifiable engineering proposition. Until the three things are built, calling a system “wise” is not rigorous — however fluent its answers.

References

AI/systems references reuse the volume' s master bibliography (checked 2026-09-19); the entries below were checked individually on 2026-09-19/20.

1. Flavell, J. H. (1979). Metacognition and Cognitive Monitoring. *American Psychologist*, 34(10), 906–911. DOI:10.1037/0003-066X.34.10.906.
2. Guo, C., Pleiss, G., Sun, Y. & Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *ICML*, PMLR 70:1321–1330. arXiv:1706.04599.
3. Chow, C. K. (1957). An Optimum Character Recognition System Using Decision Functions. *IRE Trans. Electronic Computers*, EC-6, 247–254.
4. Chow, C. K. (1970). On Optimum Recognition Error and Reject Tradeoff. *IEEE Trans. Information Theory*, IT-16 (exact pages [CITATION NEEDED]).
5. El-Yaniv, R. & Wiener, Y. (2010). On the Foundations of Noise-free Selective Classification. *JMLR*, 11, 1605–1641.
6. Geifman, Y. & El-Yaniv, R. (2017). Selective Classification for Deep Neural Networks. *NeurIPS*.
7. Geifman, Y., Uziel, G. & El-Yaniv, R. (2019). SelectiveNet. arXiv:1901.09192.
8. Angelopoulos, A. N. & Bates, S. (2021). A Gentle Introduction to Conformal Prediction. arXiv:2107.07511.
9. Kadavath, S. et al. (2022). Language Models (Mostly) Know What They Know. arXiv: 2207.05221.
10. Shinn, N. et al. (2023). Reflexion: Language Agents with Verbal Reinforcement Learning. *NeurIPS*. arXiv:2303.11366.
11. Ha, D. & Schmidhuber, J. (2018). World Models. arXiv:1803.10122.
12. Hendrycks, D. & Gimpel, K. (2017). A Baseline for Detecting Misclassified and Out-of-Distribution Examples. *ICLR*. arXiv:1610.02136 (abstract fields [CITATION NEEDED]).
13. Lewis, P. et al. (2020). Retrieval-Augmented Generation. *NeurIPS*. arXiv:2005.11401.
14. Schick, T. et al. (2023). Toolformer. *NeurIPS*. arXiv:2302.04761.
15. Bellman, R. & Åström, K. J. (1970). On Structural Identifiability. *Mathematical Biosciences* (exact pages [CITATION NEEDED]).
16. Ji, Z. et al. (2023). Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys* (exact pages [CITATION NEEDED]).
17. Castro, M. & Liskov, B. (1999). Practical Byzantine Fault Tolerance. *OSDI*.

Appendix: Provenance of the UDOS Numbers

Number	Data file	Level
World model 967,796 params; blind MSE 0.0459→0.01037; latency 2.36 ms	v7_verification.json (params / improvement_vs_v701 / latency_ms)	verified
Conformal 80/90/95: oracle 0.7839/0.8984/0.9577, blind 0.7758/0.9001/0.9559; widths in Table 1	v7_verification.json (coverage.*)	verified
Skill: $v_0 + 0.318$, $\omega + 0.426$, v_2 +0.820, acceleration -0.508 (n=480)	v7_verification.json (estimator_param_error.*)	main gate verified, breakdown cpu-proto
21 gates all pass; trigger rates 0.6341/0.6624/0.1544/0	v7_verification.json (gating)	verified
Embodied hybrid direct 75%/ motion 52.5%/hybrid 100%; collisions 0.25→0; takeovers 1.0→0.047	embodied_hybrid_demo.json (P7)	cpu-proxy
BFT n=7,f=2 all correct; market conservation 240; P3 24/24 (single seed)	P2/P3/P11 sources	cpu-proto

(End. P0c is the prologue “self-boundary” axis of the flagship trilogy; the structure axis is P0, the scale axis P0b.)